That is, $f'(x) = 0$.

or $\boxed{\frac{d}{dx}(k)=0}$.

(2) The power function $y = x^n$, $n > 0$ is an integer.

Let $f(x) = x^n$.

Then, $f(x+\Delta x) = (x+\Delta x)^n$ and

$f(x+\Delta x) - f(x) = (x+\Delta x)^n - x^n$.

This implies, $\frac{f(x+\Delta x)-f(x)}{\Delta x} = \frac{(x+\Delta x)^n - x^n}{(x+\Delta x)-x}$

This implies, $\lim\limits_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x} = \lim\limits_{\Delta x\to 0}\frac{(x+\Delta x)^n - x^n}{(x+\Delta x)-x}$

$= \lim\limits_{y\to x}\frac{y^n - x^n}{y-x} = nx^{n-1}$, where $y = x+\Delta x$ and $y \to x$ as $\Delta x \to 0$.

That is, $\frac{d}{dx}f(x) = nx^{n-1}$ or $\boxed{\frac{d}{dx}(x^n)=nx^{n-1}}$.

Corollary 10.1

When $n=\frac{p}{q},(p,q)=1$, $\frac{d}{dx}\left(x^{\frac{p}{q}}\right)=\frac{p}{q}x^{\frac{p}{q}-1}$.

Corollary 10.2

For any real number α, $\frac{d}{dx}(x^\alpha)=\alpha x^{\alpha-1}$.

For instance,

(1) $\frac{d}{dx}(5) = 0$ since 5 is a constant.

(2) $\frac{d}{dx}(x^3) = 3x^2$, by power function rule.

(3) $\frac{d}{dx}\left(x^{\frac{3}{2}}\right) = \frac{3}{2}x^{\frac{3}{2}-1}=\frac{3}{2}x^{\frac{1}{2}}$, by power function rule.

(4) $\frac{d}{dx}\left(x^{\sqrt{2}}\right) = \sqrt{2}x^{\sqrt{2}-1}$, by power function rule.

(5) $\frac{d}{dx}\left(x^{\frac{2}{3}}\right) = \frac{2}{3}x^{\frac{2}{3}-1}=\frac{2}{3}x^{\frac{-1}{3}},(x\neq 0)$, by power function rule.

(6) $\frac{d}{dx}(100x^9) = 100\frac{d}{dx}(x^9)=100\times 9x^{9-1}=900x^8$ by theorem 10.6.

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